Onur Keleş

Istanbul, Turkey (visiting UPenn, Jan – May 2027) | +90 536 872 98 81
onur.keles1@bogazici.edu.tr | Personal Website
Google Scholar | Github | LinkedIn

RESEARCH FOCUS

I bring training in human multimodal communication (gesture, sign language, gaze, and body movement) from linguistics and psycholinguistics, and use it to evaluate and build multimodal models. My work tests vision–language models against human perception of hand and body movement in video, models human movement from video with pose estimation and kinematics, and generates sign language motion for a 3D avatar from text with LLMs. I build Python and PyTorch pipelines for model training and evaluation. First-author papers at ACL 2026 (main) and two NAACL 2025 workshops.

EDUCATION

Visiting Scholar, University of Pennsylvania, Department of Linguistics Jan 2027 – May 2027 (incoming)
Host: Gareth Roberts. Dissertation research on how humans and vision–language models process visual language.

PhD in Linguistics, Boğaziçi University, Istanbul, GPA 4.0 Jan 2025 – Present (expected 06/2029)
Advisors: Kadir Gökgöz (Boğaziçi); Aslı Özyürek and Esam Ghaleb (Max Planck Institute for Psycholinguistics)

M.A. in Linguistics, Boğaziçi University, GPA 4.0 Sep 2021 – Jan 2025
Thesis: Phonetic reduction in Turkish Sign Language: kinematic evidence from pose estimation

B.A. in Foreign Language Education, Boğaziçi University, Summa Cum Laude (4.0) Aug 2017 – Jul 2021

Graduate CS/AI coursework: Human-Inspired Machine Intelligence; Generative AI & Deep Learning; Transformer-Based Pre-training in NLP; Natural Language Processing; Machine Translation; Data Science.

RESEARCH EXPERIENCE

Max Planck Institute for Psycholinguistics, *Affiliated Researcher* Jun 2024 – Present

- Built the *Visual Iconicity Challenge*, a video benchmark of 96 signs (Sign Language of the Netherlands) that tests whether vision–language models perceive visual form and meaning as humans do: phonological form prediction, meaning inference from form, and graded iconicity ratings, with deaf and hearing human baselines.
- Evaluated 17 models: 11 open models from 4B to 72B (Qwen2.5-VL, Qwen3-VL, VideoLLaMA2, LLaVA-OneVision, Gemma 3, MiMo-VL) on up to four A100 GPUs in zero- and 4-shot settings, and 6 closed models (GPT-5.4, Gemini 3.1 Pro, and others) through APIs. Code.
- Models follow human difficulty patterns (handshape harder than location) but mostly fail to infer meaning from form and prefer object-based over action-based signs, unlike humans (ACL 2026).
- Follow-up (HSP 2026): adding visual context of agent–object relations improved models' form–meaning mapping, but a large gap to human performance remained.

Human movement from video: pose estimation and kinematics (M.A. thesis) 2023 – Present

- Tracked hand and upper-body landmarks with MediaPipe in 23,564 video frames of 28 deaf signers retelling cartoon stories; filtered and normalized the trajectories and computed sign duration, hand distance, and signing space size (Python, OpenCV, NumPy).
- Mixed-effects models (lme4) showed that signs are longer, move more, and use more space when a referent is new, and shrink gradually as it becomes more predictable (under review, Glossa).
- Age of sign language acquisition did not change sign kinematics, but early signers told events more briefly.

Sign language generation: Turkish text to TİD (with M. İnan, Northeastern & Amazon) Ongoing

- Fine-tuned open LLMs (Gemma 3, Qwen 3) and prompted closed LLMs to translate Turkish into TİD glosses; linguistic modules stitch dictionary clips into continuous pose sequences that drive a 3D avatar.
- Larger fine-tuned open models beat closed models on BLEU, chrF, and ROUGE; evaluated with Deaf and expert signers.

LLaMA-2-Econ: LLM fine-tuning 2024

- Fine-tuned LLaMA-2-7B with QLoRA/PEFT; it beat GPT-3.5 and GPT-4 on abstract classification.

Boğaziçi University, Psycholinguistics Lab Jun 2022 – Present

- *MultiplEYE* (EU COST Action): related eye movements of 57 readers to surprisal from six LMs at three context lengths with mixed-effects models; the current sentence mattered most, earlier sentences added no significant gain.
- Compared human self-paced reading times with surprisal and attention patterns of Turkish LMs (GPT-2, LLaMA-3); larger models relied more on word order, smaller ones on plausibility (CMCL @ NAACL 2025).

SELECTED PUBLICATIONS

1. **Keleş, O.**, Özyürek, A., Ortega, G., Gökgöz, K., & Ghaleb, E. (2026). The Visual Iconicity Challenge: Evaluating Vision–Language Models on Sign Language Form–Meaning Mapping. *ACL 2026 (Main, Long Papers)*.
2. **Keleş, O.**, & Dinçtopal Deniz, N. (2025). When men bite dogs: Testing good-enough parsing in Turkish with humans and large language models. *CMCL Workshop @ NAACL 2025*.
3. **Keleş, O.**, Günay, B., & Doğan, B. (2025). Cognate and contact-induced transfer learning for Hamshentsnag: A low-resource and endangered language. *LM4UC Workshop @ NAACL 2025*.
4. **Keleş, O.** & Bayraklı, Ö. T. (2024). LLaMA-2-Econ: Enhancing title generation, abstract classification, and academic Q&A in economic research. *FinNLP Workshop @ LREC-COLING 2024*.
5. **Keleş, O.** & Dinçtopal Deniz, N. (2024). A comparative study with human data: Do LLMs have superficial language processing? *IEEE SIU 2024*.

Under review

1. **Keleş, O.**, Inan, M., & Gökgöz, K. Gloss, Stitch, Sign: Low-resource sign language generation for Turkish Sign Language with LLMs and a community dictionary. *ACL Rolling Review (Oct 2026)*.
2. **Keleş, O.**, Dinçtopal Deniz, N., & Gökgöz, K. Phonetic reduction as a function of referential accessibility: Kinematic evidence from Turkish Sign Language. *Glossa: a journal of general linguistics* (Q1).
3. **Keleş, O.**, Kayhan, S. A., Bektaş Yüksel, S., & Dinçtopal Deniz, N. How much context does LM surprisal need to predict reading times? Evidence from Turkish eye tracking. *ACL Rolling Review (Oct 2026)*.

Journals: *Journal of Pragmatics* (2023, Q1); *Language Acquisition* (2022, Q1, Best Article Award). **Talks:** invited talk, Max Planck Institute for Psycholinguistics (2025); posters at HSP 2025, HSP 2026, AMLaP 2025.

TECHNICAL SKILLS

Programming	Python (PyTorch, NumPy, OpenCV), R, Git/GitHub, LaTeX
ML & models	Hugging Face Transformers, QLoRA/PEFT fine-tuning, multi-GPU inference on an A100 HPC cluster, vision–language model and LLM evaluation, OpenAI and Gemini APIs (Vertex AI)
Multimodal communication	Sign language and gesture analysis, video annotation (ELAN), pose estimation (MediaPipe), eye-tracking, online perception experiments (PCIbex)
Modeling & statistics	Experimental design; mixed-effects regression (lme4), polynomial models, held-out model comparison; correlation with human judgments; item-level error analysis
Evaluation	Benchmark design, zero- and few-shot prompting, BLEU, chrF, ROUGE, F1
Languages	Turkish (native) · English (C2) · Spanish (B1) · Turkish Sign Language (research experience)

INDUSTRY & TEACHING

micro1 & Mercor, *AI Training Expert and Evaluator (freelance)* 2025 – Present
Multimodal annotation (speaker diarization, audio-visual grounding) and LLM response evaluation for frontier model training.

Codeswitch Software, *Research Intern* Jan 2021 – Aug 2021
Sign language learning application, supported by a TÜBİTAK STAR research fellowship.

Boğaziçi University, *Research and Teaching Assistant, Linguistics* Sep 2022 – Present
Computational Linguistics with Python; Linguistic Methodology with R; Morphology; Turkish grammar.

AWARDS & SERVICE

Awards	AlgoRun Hackathon winner (2024); Best Paper, Akademik Bilişim (2024); ACL 2026 D&I Subsidy.
Service	Reviewer, NeurIPS; In-person volunteer, ACL 2026 (San Diego).